

Meeting 1022

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資料介紹

SECOM

▶ 樣本數 1567

▶ 變數 Time / x1 / x2 / ... / x590 / PassFail

■ Features: 591, Y: PassFail (binary classification)

■ Time: 2008-07-19 11:55:00

■ PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)

■ Missing values: 41951

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文獻回顧 I

► Lai (2002)

1 方法

- 1 Data preprocessing
- 2 KNN-imputation
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- 4 imbalance - SMOTE to 1:1

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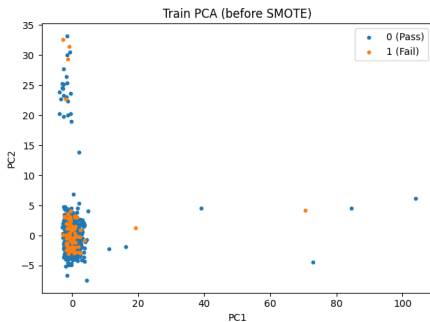
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► Lai (2002)

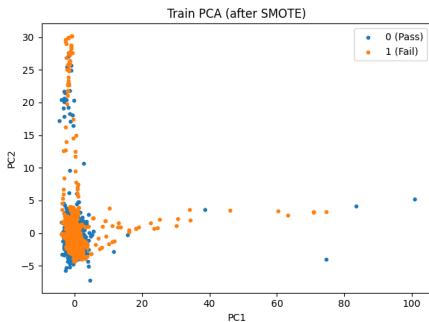
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文獻回顧 I-方法 (SMOTE)



SMOTE before



SMOTE after

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1 方法

- 1 Data preprocessing
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- 5 0.8 training, 0.2 testing by decision tree

2 結果

- 1 Best: ACC 86.94%, FAR 9.22%
- 2 反查 Fail 節點

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研究動機

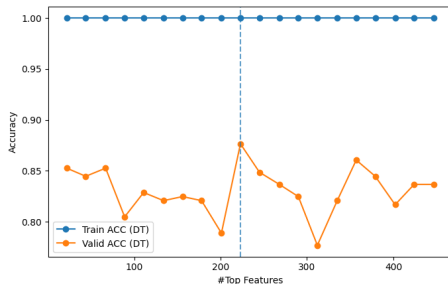
- ▶ K-folds cross validation
 - Split to training&testing first
 - Robust performance
- ▶ Metric
 - ACC, FAR, Sensitivity, Specificity
- ▶ $y = \hat{f}(x)$
 - Important features

研究結果 (Decision Tree (Test) $k = 223$)

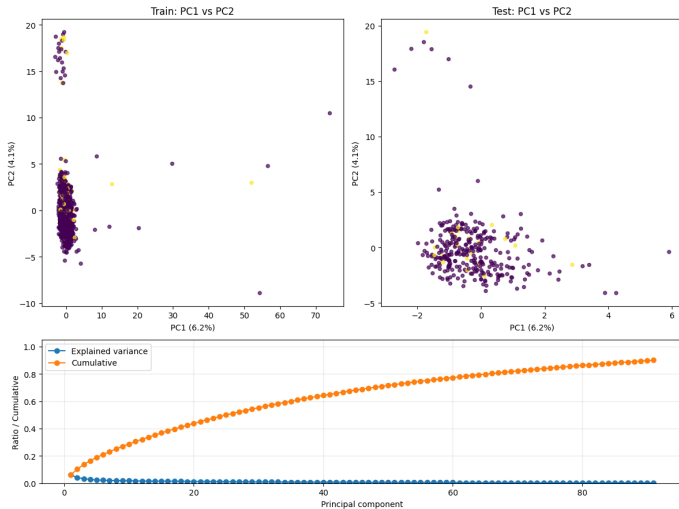
Accuracy 0.8662
FAR 0.0922
Sensitivity 0.2857
Specificity 0.9078
GM 0.5093

Confusion Matrix

	Pred 0	Pred 1
True 0	266	27
True 1	15	6



研究結果



參考資料



Li-Fu Lai (2022)。

使用 KNN 和決策樹方法對填補缺失值的製造數據進行失效預測和原因
追蹤分析

The End

Questions & Comments